HerbCheck: a herb–medication interaction reference

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Problem

Many people take herbal supplements alongside prescription drugs and rarely tell their pharmacist. Interaction information is scattered, and typos or alternate names make quick lookups unreliable.

Solution

HerbCheck lets a user enter herbs and medications and returns a plain-language list of known interaction flags. Each flag shows its attention level, mechanism, evidence type and a confidence score.

Key features

- **Safe by design.** Unknown names are only suggested (“did you mean…”), never auto-accepted. A missing record is never reported as safe.
- **Confidence gate.** Findings with weak or mixed evidence send the whole result to “needs review”.
- **Semantic matching.** Vector search in IBM Db2 (VECTOR_DISTANCE) handles typos and alternate names.
- **Evidence retrieval.** LangChain retrieves the supporting passage for each finding.
- **Audit trail.** Each check is logged as counts and a digest only, with no personal data.
- **Optional AI explanation.** A grounded summary limited to the retrieved facts; a template is used if no model is configured.

How it works

Input is normalized and resolved with Db2 SQL plus vector search. A Db2 join finds interactions, retrieval attaches the evidence, and a gate decides the status. Then an explanation is generated and the check is audited. Stack: IBM Db2 12.1, LangChain, FastAPI, fastembed (384-dimension embeddings), with an optional n8n front door.

Impact and limits

HerbCheck shows how a reference tool can fail safely: it declines to guess, flags weak evidence and never equates “not found” with “safe”. It is an educational research prototype built on a small unreviewed starter set (25 herbs, 160 drugs, 44 rules) using synthetic reference data only. It is not medical advice or a medical device. Next steps are clinician-reviewed content and more herbs and sources.

Code: github.com/udaytx009/uday-shankar-herbcheck-db2 (MIT license)